THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR PRE-MOCK EXAMINATION

031/1 **PHYSICS 1**

Time 2:30 Hours **March, 2025**

Instructions

1. This paper consists of sections **A**, **B** and **C** with total of **eleven** (**11**) questions
2. Answer all questions in section **A** and **B** and any **two (2)** questions from section **C**.
3. Communication devices and any unauthorized materials are **not** allowed in the examination room.
4. Non programmable calculator and mathematical table may be used.
5. Write your **Examination Number** on every page on your answer booklet(s)
6. Where necessary the following constants may be used:

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(i) Acceleration due to gravity g = 10m/s^2

(ii) Specific heat capacity of water = 4200Jkg^{-1}K^{-1}

(iii) Speed of light in air = $3\times10^8 ms^{-1}$

(iv) Speed of sound in air = $330 ms^{-1}$

(v) Linear expansivity of iron = 0.000012 per ^oC

(vi) Linear expansivity of brass = 0.000019 per ^oC

(vii) Pie (π) = 3.14

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number. **(10 marks)**

(i) A density bottle has mass of 70g when empty, 90g when full of water and 94g when full of a liquid. What is the relative density of the liquid?

A. 0.34 B. 0.83 C. 0.96 D. 1.04 E. 1.2

(ii) A force of 40N is necessary to extend a spring by 20mm. If a mass of 0.5kg is attached to the lower end of the spring, what will be the extension of the spring in mm?

A. 0.20 B. 1.00 C. 2.50 D. 5.00 E. 10.00

(iii) A measuring cylinder is filled with oil. A girl says that the pressure at the bottom of the cylinder depends on

1. the height of the oil
2. the area of the bottom of the cylinder
3. the density of the oil

Which of A, B, C, D, E is correct?

A. 1, 2 and 3 B. 2 and 3 C. 1 and 2 D. 1 and 3 E. 1 only

(iv) Which statant is correct about charging by induction?

A. A conductor is rubbed with an insulator
B. A charge is produced by friction
C. Negative and positive charges are separated
D. A positive charge induces a positive charge
E. Electrons are sprayed into an object

(v) A student wants to test a piece of metal to determine whether it was a magnet or not. What observation should be taken on the metal?

A. Attracted steel filings
B. Repel a metal bar
C. Attracted a magnet
D. Repelled a known magnet
E. Remain stationary

(vi) A mass is projected upwards with a velocity of 10m/s, what is the maximum height reached in metre?

A. 1 B. 5 C. 10 D. 20 E. 100

(vii) At what height above the ground must a mass of 5kg be to have a potential energy equal in value to the kinetic energy possessed this mass with a velocity of 10m/s?

A. 1m B. 5m C. 10m D. 50m E. 100m

(viii) Which of these resources of energy is non-renewable?

A. Wave energy B. Biofuels C. Radiant energy D. Fossil fuel
E. Geothermal energy

(ix) A typist uses a new carbon paper under her top typing paper for making a copy of a letter. When she holds the carbon paper close to a plane mirror, she can read the letter. This is because the mirror;

A. forms an image the same size as object
B produces an inverted image
C produces a laterally image
D forms a virtual image
E forms an image behind the mirror.

(x) Which of the following is (are) true both for light waves and for sound waves;

1. They can travel through the vacuum.
2. Speed of wave $= frequency \times wavelength$
3. They transfer energy from one place to another

A. 1, 2, 3 B.1,2 C. 2, 3 D. 1 only C. 3 only

2. The **figure 1** in list B shows an electromagnetic waves spectrum (not drawn in a scale) labeled by alphabets. Match the item in **list A** with the letter of the correct response in the diagram in **list B**. Write the letter beside the item number. **(6 marks)**

LIST A	LIST B
(i) The value of velocity ratio increases with increase the length of the plunk.	A. Wheelbarrow
(ii) First class lever	B. Claw bar
(iii) Operates with the principle of pressure in fluid.	C. Pulley
(iv) An example of third-class lever	D. Wheel and axle
(v) It can be used to move a load from one point to another horizontally.	E. Inclined plane
(vi) Used in lifting water in a bucket from a deep well.	F. Hydraulic press
	G. Gear
	H. Screw jack
	I. Shovel

SECTION B (54 Marks)

Answer **all** questions from this section

3. (a) A body dipped in a liquid experience an upthrust. Explain three factors on which the upthrust depends. **(3 marks)**

(b) Two boys, A and B are carrying an electric pole of weight 10,000N. If A holds the pole 5m from its centre of gravity and B holds 4m from its centre of gravity. Calculate the load that each of them support. **(6 marks)**

4. (a) The efficiency of a press is 75%. If the radius of the load piston is 3cm while that of the effort piston is 1.5cm. calculate the mechanical advantage of the press. **(4 marks)**

(b) A boy with a mass 50kg jumps out of a rowing boat of mass 300kg on to the bank with horizontal velocity of 3m/s. with what velocity does the boat begin to moves backwards? **(5 marks)**

5. (a) You are provided with three mirrors; concave mirror, convex mirror and plane mirror. Which mirror will you prefer to be a sight mirror? Give two reasons. **(4 marks)**

(b) A concave mirror is used to form an image of the pin object. By using a well labeled diagram; show how you can obtain an upright and enlarged image. **(5 marks)**

6. (a) The **figure 1** shows a car pulling a wagon carrying boxes. Explain operations of friction observed in this picture. Give four points **(4 marks)**

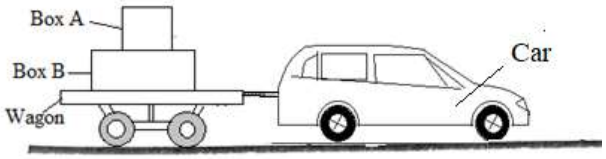


Figure 1

(b) A mass of 3kg hangs at the end of the string. Using a sketched diagram, find the horizontal forces needed to pull sideways until the string is at 30° to the vertical **(5 marks)**

7. (a) The **figure 2** shows a bimetallic thermostat used to regulate a cooler and heater in a class room to keep the temperature constant. Which of the two devices A or B should be the heater? Explain your answer. **(4 marks)**

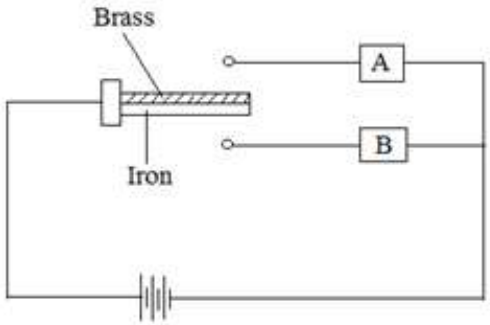


Figure 2

(b) Two beakers are of identical size and shape; one beaker is painted black and the other beaker is painted white. The beakers are filled with boiling water then covered. In which beaker will the water cool most quickly? Give a reason. **(5 marks)**

8. (a) When you take a bottle of cold soft drinks from a refrigerator, water appears on its surface. Explain this observation. **(4 marks)**

(b) An insulated cup holds 0.3kg of water at 0°C. If 0.2kg of boiling water at standard pressure is poured into the cup. What will be the final temperature of the mixture at thermal equilibrium? **(5 marks)**

SECTION C (30 Marks)

Answer **two (2)** questions from this Section

9. (a) Draw a circuit diagram which can be used to verify ohm's law in the laboratory. **(5 marks)**

(b) You are provided with a bundle of a resistance wire labeled 0.38mm diameter and resistivity of $4.9\times10^{-7}\,\Omega m$. How will you design a resistance of 2Ω using this wire? **(5 marks)**

(c) You are provided with fuse wires labeled 2A, 3A, 4A, 6A and 10A. Which of these is most suitable for two electric devices rated (240V, 200W) and (220V, 1.0kW) to operate at the same time? Justify your answer. **(5 marks)**

10. (a) What is the relationship connecting the frequency of sound source with its wavelength and the speed of sound in air? **(3 marks)**

(b) If 5s elapse between a lightning flash and the clap of thunder, how far away is the storm? **(5 marks)**

(c) A string is set vibrating by plucking at a point one-quarter of the length from the end.

(i) Draw a sketched diagram showing nodes and antinodes in a vibrating string.

(3 marks)

(ii) Write the expression of the frequency note produced by the vibrating string.

(4 marks)

11. (a) A group of form four students designed a d.c generator in their project. Unfortunately, the designed generator produces the electric power of low voltage. Suggest two alterations which can be done in order to improve the production of high voltage electric power. **(4 marks)**

(b) Three parallel wires, A, B and C of the same length are placed equidistant from one another as shown in **figure 3**. Wires A and C are fixed but wire B is free to move. If the same current *I*, is passed through each of the wires at the same time, in which direction will the wire B tend to move? Explain your answer **(5 marks)**

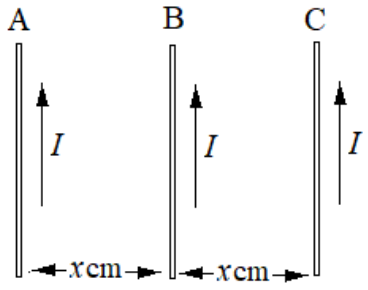


Figure 3

(c) A transformer is designed to work from 240V a.c mains. The primary coil has 4800turns. About how many turns of secondary coil would expect it to have in order to give a supply 8V to ring house bell? **(6 marks)**